Chapter0

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1 Motivation

Footwear examination is important to forensic investigation. The features provided by footwear prints are known as class characteristics (@fig-classes): they may be shared by many shoes of the same make and model, but are useful for narrowing down a subject pool and providing some association between a suspect and the scene[1]. Collection and processing of footwear data require large amounts of human labor, hence, automating the the process of identification class characteristics is essential. With advances in neural network-based vision models, automated process has become feasible.

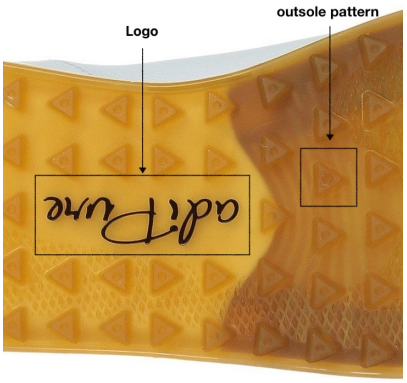


Figure 1: Example of Footwear Class characteristics

2 Neural Network Vision Models

Modern neural network-based vision models are mainly built on two categories of architectures: convolutional neural networks and attention-based networks. Although they use different mechanisms, these architectures are used as backbones to extract features from images. These

features can be used by downstream task-specific modules, which are also referred to as heads. Task-specific heads are selected depending on the objective of the model. For example, in an image classification task, a classification head is attached to the backbone to predict image-level categories. In an object detection task, detection heads are used to predict both object classes and bounding box locations. A diagram of a neural network is shown in Figure 2.

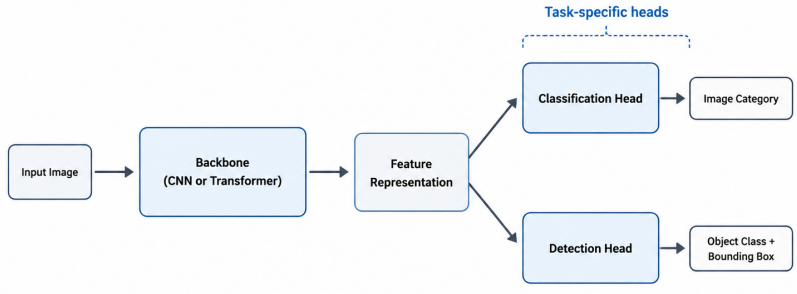


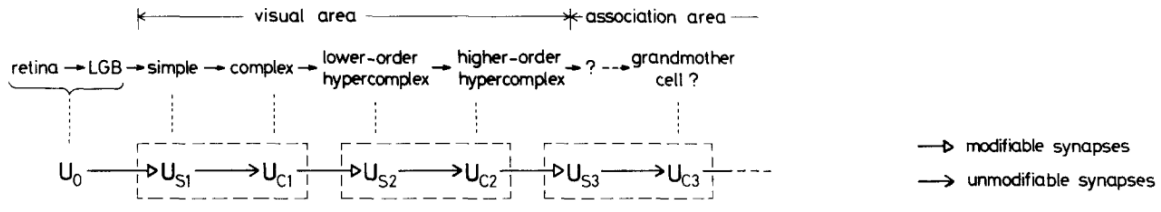
Figure 2: Vision model diagram

2.1 Convolutional-based Vision Models

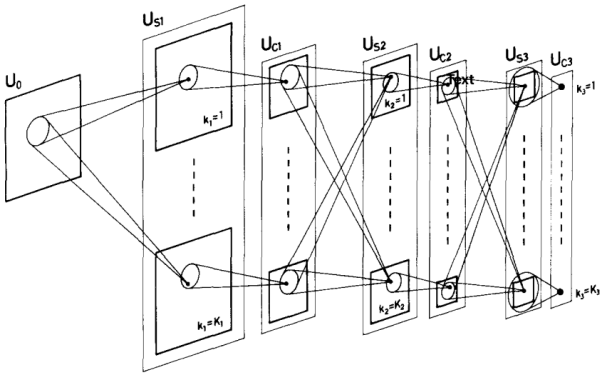
Convolutional neural networks (CNNs) are one of the most widely used neural network architectures. The development of convolutional neural networks can be traced to biologically inspired models of vision.

A hierarchy model[2] [3] proposed that the neural network in the visual cortex are structured in a hierarchical way: responses from low level cells are integrated by higher-level cells, so that more complex visual features of larger receptive field can be detected with great tolerance to positional shifts. Based on this idea, Fukushima proposed the Neocognitron[4], an early neural network model for visual pattern recognition. Figure 3 shows how the Neocognitron structure

corresponding to the biological hierarchy visual model. In the Neocognitron, the (U_s) and (U_c) layers provide an artificial analogy to simple and complex cells. S-cell layers detect local visual feature, while C-cell layers combine nearby responses. A complex visual representation is built by repeating the S-cell followed by C-cell structure.



(a) circle jittering



(b) RF5 jittering

Neocognitron[4]

Figure 3: Neocognitron design is inspired by the visual biological structure

While the Neocognitron is considered as a biologically inspired precursor, it could not be trained as an integrated end-to-end system. LeNet[5], proposed by LeCun, represented an early form of modern convolutional neural networks that could be trained end-to-end. LeNet established a basic structure for later CNN architectures and enabled image features to be learned directly from image data. In this structure, convolutional filters (Figure 4) are applied to local regions of the input image to detect spatial patterns.

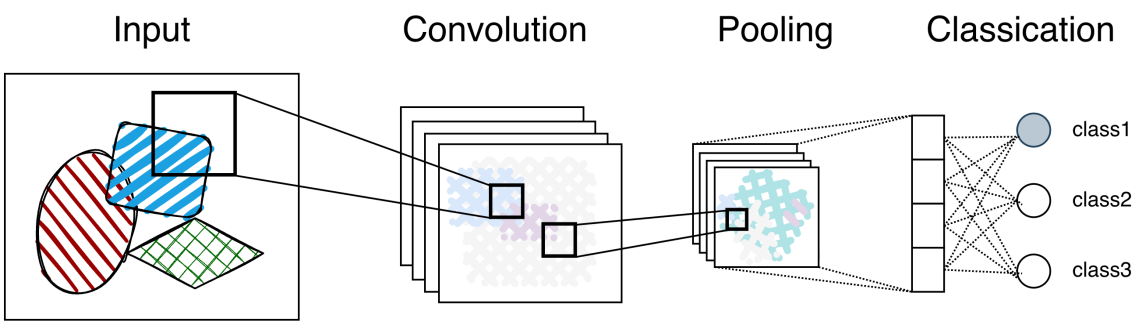


Figure 4: Convolution operations EXPLANATIONS HERE???

As a filter moves across the image, it computes responses to local visual patterns and produces a feature map that represents the strength of the filter response at different spatial locations. As the filter slides across the image, its weight will remain unchanged, these reused parameters are referred to as shared weights. The filter plays a role similar to S-cells in the Neocognitron, in that they respond to local visual patterns and provide the basis for progressively more complex feature representations. Although primary features like edges and angles are meaningful to humans, high-level features extracted by CNNs are not easily interpretable because they encode complex combination of visual patterns.

The use of local convolutional filters and shared weights gives CNNs strong image-specific inductive biases. Local convolutional filters implies the assumption that nearby pixels are more likely to be related than distant pixels. Shared weights allow the same filter to be reused across spatial locations, this design supports translation equivariance, in which a spatial shift in the input tends to produce a corresponding shift in the feature response. By stacking multiple convolutional layers, CNNs can gradually integrate local patterns into more complex and abstract visual representations. This local-to-global organization is conceptually related

to hierarchical processing in the biological visual system.

2.2 Attention-based Vision Models

Attention-based vision models are another major class of neural network-based models. Attention mechanism is introduce in a natural language processing model named Transformer[6]. The attention mechanism allows a model to assign different weights to different parts of the input. Heavier weights indicate relevant parts contributing more to the learned representation, which directly models global context. This idea was later adapted to vision modeling through the Vision Transformer[7] (ViT). The Vision Transformer inherits the attention mechanism to model region relevance. An input image is represented as a sequence of segmented visual patches, and the patches are assigned attention weights according to their relevance to the representation. In this way, the model learns relationships among different patches and captures long-range dependencies across the image. This differs from CNNs, which primarily learn local patterns through convolutional operations and gradually build broader spatial context though stacking layers.

The diagram is shown in Figure 5 The input image is first split into image patches. These patches are embedded as tokens and then processed by the self-attention module. Self-attention assigns different weights to different patches according to their relevance. The output attention maps show the spatial distribution of these attention weights across the image.

While CNNs assumes that local regions are more likely to be related by employing local filters, the Vision Transformer does not impose receptive fields and models global context with a

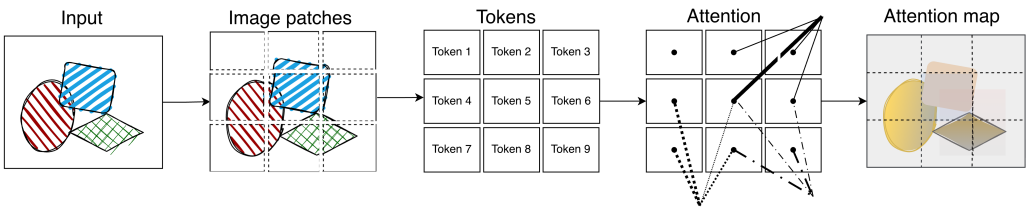


Figure 5: Vision Transformer

direct mechanism. Therefore, the Vision Transformer often requires large-scale training data to learn robust visual representations.

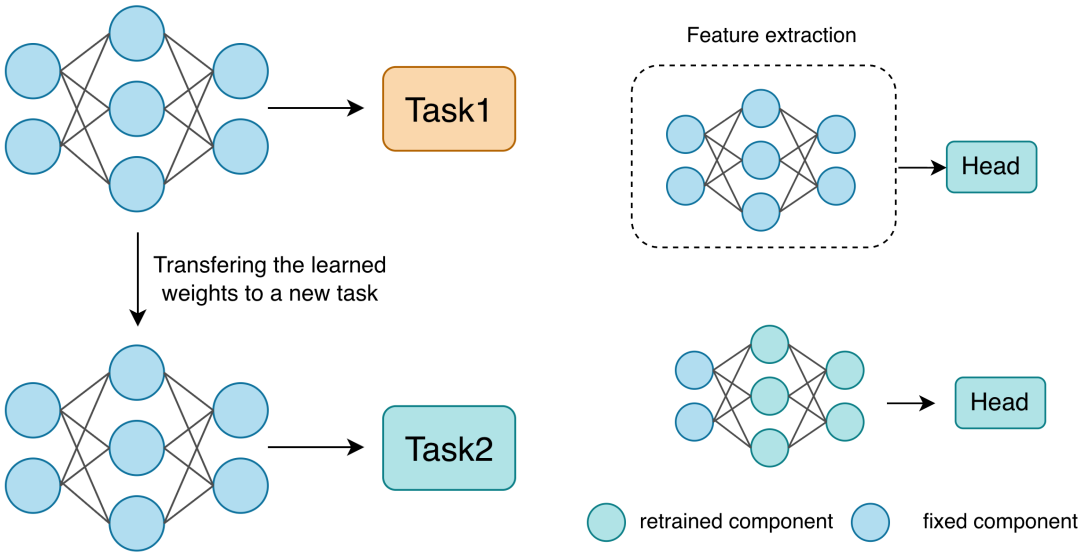
2.3 Heads and Transfer Learning

Both CNN-based and attention-based architectures can be used to extract image features. However, these features must be used by a task-specific head to complete a particular task. For image classification, a classification head maps the features to category predictions. For object detection, detection heads use the feature representation to predict both object categories and locations.

In a specific project, the task-specific head is usually selected or modified according to the objective of the study. For example, in an image classification task with seven categories, the final classification head need to be replaced by a seven-class classifier. The model is then trained by minimizing a task-specific loss function, such as a classification loss for category prediction.

In many applied vision projects, the computing resource or the number of project-specific training images are limited. Therefore, transfer learning is a commonly used strategy in the

training to improve efficiency and performance. In transfer learning, a model is first pretrained on a large scale dataset, such as ImageNet[8] for image classification, and then adapted to the target dataset. Through pretraining, the backbone learns general visual features, such as edges, textures, and shapes, are considered transferable across different image datasets and vision tasks[9]. Then the task-specific head can be replaced or fine-tuned for the new task. The task-specific model is further trained using the target data. During the training, the pretrained backbones can be freezed or fine-tuned together with the new head, because it was found that features in deeper layers become increasingly specific to the original task[9]. The transfer learning diagram is shown in Figure **??**.



(a) Transfer learning

(b) Training

Previous studies have demonstrated that transfer learning is effective, especially across visual network architectures and tasks[7] [10][11] [12].

The goal of the present study is to locate and classify footwear class characteristics. These

characteristics often consist of geometric patterns that contain both local features and broader spatial context. For example, recognizing a circular pattern requires the model to identify local curvature as well as the overall closed shape. Therefore, both CNN-based and attention-based models are considered suitable for this task. Given the limited amount of labeled training data, transfer learning is employed to improve model performance.

3 Image Classification

Image classification is a fundamental task in computer vision because it provides a basic framework for learning and evaluating visual feature representations. A classification dataset consists of images and their corresponding class labels, which are assigned to the entire image. In a classification model, a classification head uses these features that output from a backbone to predict the image category. Many widely used backbones were originally developed and evaluated on image classification datasets. The representations learned through classification later became the foundation for models used in various vision tasks. This section reviews four representative architectures: AlexNet[13], VGG[11], ResNet[12], and the Vision Transformer[7]. These architectures illustrate important developments in convolutional and attention-based visual modeling.

3.1 Data

An image classification dataset consists of images and their labels. In the simplest case of single label task, each image is assigned to one category. The labeled images are divided into training, validation, and test sets. The training set is used to optimize the model parameters by minimizing a classification loss function, which measures the difference between the predicted class and observed labels. The validation set is used to identify potential overfitting, and the test set consists of held out images which are reserved for the final evaluation. During training, both the backbone and the classification head can be optimized: the backbone learns visual feature representations, and the classification head maps these representations to class predictions.

3.2 Architectures

3.2.1 AlexNet

AlexNet, introduced by Krizhevsky[13], was an influential convolutional neural network. It demonstrated that a large convolutional network could be successfully trained on a large-scale image dataset, and hierarchical visual features could be learned directly from image data.

The success helps to establish deep CNNs as a central approach in computer vision. Its architecture incorporated several techniques that facilitated large-scale CNN training to allow effective robust feature representation learning and large scale image classification.

Although transfer learning had been studied earlier than AlexNet, the success of AlexNet contributed to the prevalence of transfer learning. Fine-tuning a deep neural network initialized with pretrained weights, such as those from AlexNet, has been shown to outperform training the same network from random initialization[9]. Transfer learning with networks pretrained on large-scale datasets therefore provides a general paradigm for various vision-related tasks.

Another practice popularized by AlexNet was data augmentation. Data augmentation artificially enlarges the training dataset using label-preserving transformations, which helps the network learn more general features and reduces overfitting. AlexNet applied several augmentation techniques (Figure 7), including random cropping, horizontal reflection, and RGB intensity perturbation. These operations do not change the semantic content of the image but expose the model to different variations of the same object, improving its ability to generalize to unseen data.

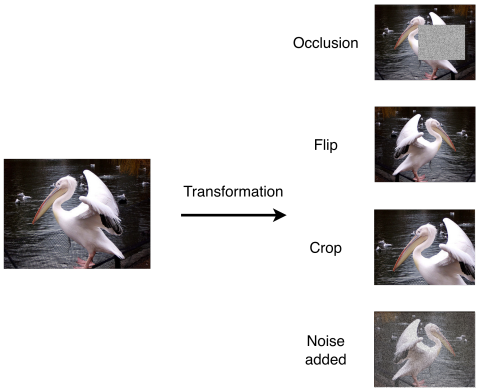


Figure 7: Data augmentation[14]

3.2.2 VGG

Building on the success of AlexNet, deeper networks was proposed to improve feature learning, representatives of the deep networks are the configurations proposed by Simonyan et al.[11]. These configurations became widely know as VGG networks, named after the Visual Geometry Group at the University of Oxford. The two commonly known architectures are VGG-16 and VGG-19, which use a simple and uniform design based on repeated small convolutional filters (3 by 3) and contain 16 and 19 learnable layers, respectively. Compared with AlexNet, which contains eight learnable layers, VGG substantially increased network depth, see Figure 8. The study demonstrated that this deeper architecture improved image classification performance, suggesting an improved ability to learn discriminative visual representations.

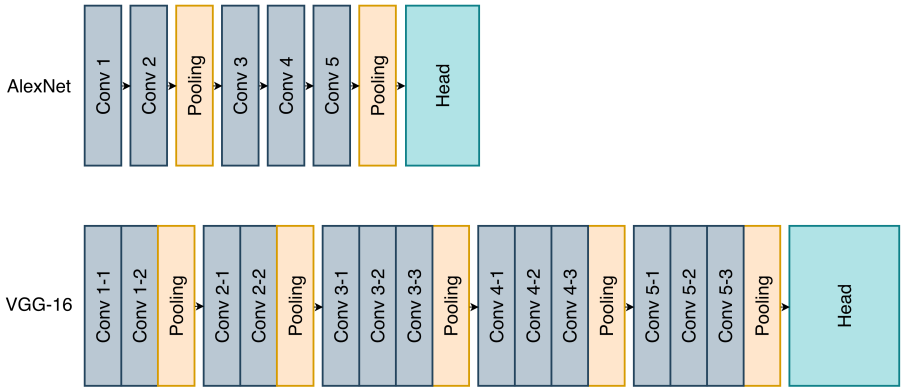


Figure 8: AlexNet: 8 learnable layers, VGG-16: 16 learnable layers

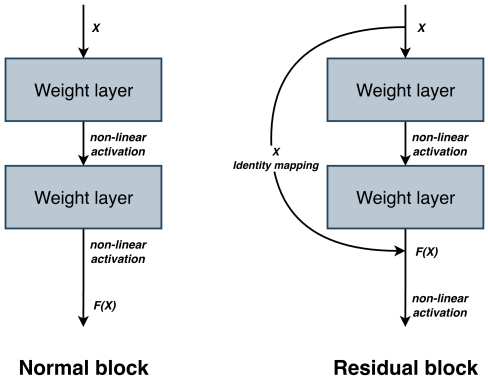
3.2.3 ResNet

The deep layer design of CNNs brought an unexpected degradation despite better performance. It was found that adding more identical mapping layers could result in higher training error as

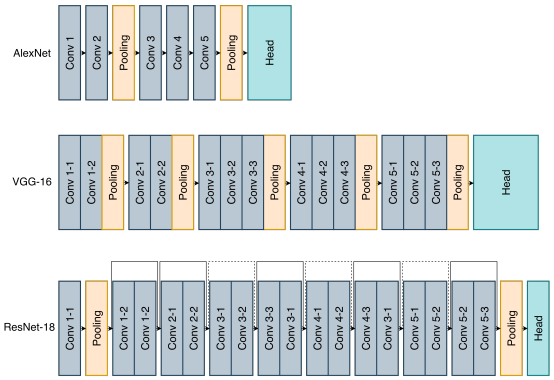
well as test error[15]. This observation suggests that the performance degradation is primarily associated with optimization difficulties rather than overfitting.

ResNet architectures[12] addressed the problem by introducing a deep residual learning module. This module adds a shortcut connection to add the input directly to the output of a few layers. The design makes optimization easier by learning the residual modification relative to the input rather than the full mapping. If the modification is small, as in an identity mapping, the optimization becomes much easier than in the normal design (Figure 9a).

ResNet networks has five standard variants, ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152. Although ResNet-18 is the shallowest variant, it still has 18 learnable layers, which is deeper than the VGG-16 (Figure 9b). ResNet achieved improved image classification performance while enabling much deeper networks to be trained effectively.



(a) Residual block and normal block



(b) Learnable depth: AlexNet (8 layers), VGG-16 (16 layers), and ResNet-18 (18 layers)

3.2.4 ViT

While CNNs rely primarily on local convolutional operations to extract visual features, the Vision Transforner (ViT[7]) introduced a different approach that captures global dependencies across an image. Attention weights are learned to represent relationships among image patches. Unlike CNNs, which have inductive biases such as locality and translation equivariance built into the architectures, ViT uses global self-attention and imposes relatively little two-dimensional neighborhood structure. Therefore, ViT performance can be more sensitive to the amount of project-specific data

These architectures were initially proposed as classification models, which includes feature extraction and class prediction. The major architectural difference lie in the feature extraction designs (backbone) rather than the classification heads. The classification heads serve a similar role to map extracted features to class predictions.

Since backbone reflects the main architectual characteristics of models designed for the same task, architecture names are commonly used to describe the unique underlying feature extraction design when these models are adapted as backbones for other tasks. For example, the same detection framework can be implemented with different backbones, such as VGG or ResNet, and these variants are often referred to by their backbone architectures.

!!!!!!!!!!!!!!!!!!!!!!!!!!!! Representative backbones options for visual feature learning, and it is therefore worthwhile to evaluate their performance in classifying footwear characteristics. !!!!!!!!!!!!!!!!!!!!!!!!

4 CNN-based Object Detection

To achieve the goal of the present study, footwear characteristics must be both localized and classified to automate the process. Therefore, the task is formulated as an object detection problem rather than an image classification problem.

4.1 Object Detection Designs

In the detection workflow, the input image is first processed by a visual backbone to extract feature representations. A detection head then uses these representations to locate and classify target patterns. Object detection models can be broadly categorized into two-stage and one-stage models. Two-stage detectors first generate candidate regions that may contain objects and then classify and refine these regions, multiple networks are trained at the same time. Faster R-CNN is a representative two-stage detector (Figure 10 upper panel). In contrast, one-stage detectors predict object locations and class labels directly from image features without a separate region proposal stage (Figure 10 lower panel). YOLO models are representative examples of this approach.

One-stage detectors simplified the architecture by excluding the additional region proposal stage, this design reduces computational cost and enables faster inference, making them suitable for real-time applications. However, without a separate proposal and refinement process, one-stage detectors may have relatively lower localization accuracy compared with two-stage detectors. The present study does not require real-time detection and prioritizes accurate

localization, Faster R-CNN was selected as the detector.

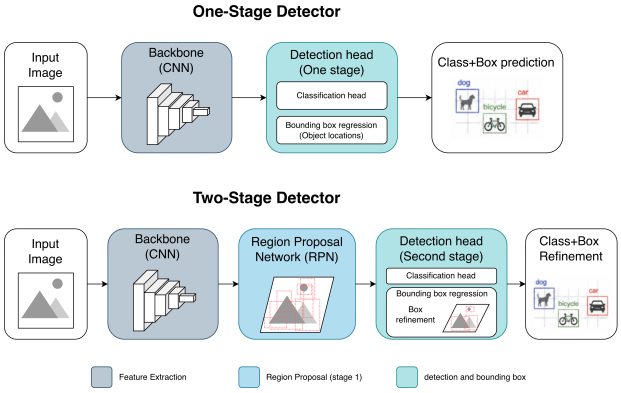


Figure 10: One stage YOLO and two stage FasterRCNN

4.2 Faster R-CNN Backbones and Heads

Faster R-CNN framework consisting of a backbone, a Region Proposal Network (RPN), and a second-stage detection head. The backbone is responsible for converting the input image into feature representations that retain visual and spatial information in the image. Based on the information, the RPN generates candidate regions that possibly contain the objects. The candidate regions will be send into the detection head. The detection head has two parallel branches: a classifier that predicts categories for the regions, and a regressor that adjusts the coordinates of the proposed box.

Faster R-CNN implementations can be realized by changing the backbone while the overall framework remains unchanged. As mentioned previously, backbone designs fundamentally affect the feature representations provided to the detector, and consequently, detection performance. Different backbone designs can provide different levels of semantic abstraction and

retained spatial information[16]. The retained spatial resolution is particularly important for localization because it determines how much positional detail is preserved.

4.3 Data

Object detection has a different data structure than the classification task. To obtain the ability of predicting spatial location, the training data must provide positional information in addition to the class label. Input images are paired with these human-provided annotations, which serve as the ground truth that guides model training and evaluation. For example, in Figure 11, the image is annotated with both bounding boxes to indicate spatial locations as well as class labels for objects in the bounding boxes. An individual object detection image allows multiple target instances, each instance associates with a class label and a spatial annotation. The annotated regions can belong to the different classes, and it is common to find them overlap.

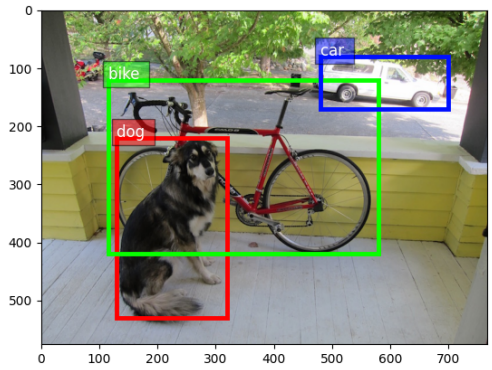


Figure 11: Object detection data[14]

Low quality input would limit the network performance, hence, annotation quality is an impor-

tant consideration in object detection. Previous studies have shown that small discrepancies in location annotation, such as different box sizes, and noisy can influence localization performance, especially for small objects[17] because its susceptibility to errors. In addition, deviation from intended localization has been shown to reduce the detection performance of Faster R-CNN[18].

5 Relevance to the Present Study

The footwear feature automation can be framed as an object detection task that consists of two main stages: first locating footwear class characteristics or patterns, and classifying the detected regions. However, approaching the problem is more than adapting the network architecture.

5.1 Challenge

Object detection requires precisely annotated training data so that the network can learn both spatial and categorical information. Previous studies have reported that degraded annotation quality can significantly reduce detection performance[17,18]. Therefore, annotation quality is an important factor in network performance.

Unlike general-purpose pretraining datasets, where object classes are typically clearly defined, footwear features can be more complex and ambiguous. For example, Figure 12a shows a feature coded as a "circle" in the forensic footwear tool SoleMate. However, the feature itself

consists of four individual arcs. It is therefore open to discuss whether this feature should be annotated as one bounding region labeled as a circle or as four separate regions labeled as arcs.

In addition to this ambiguity in grouping individual elements, uncertainty can also arise from unclear boundaries between feature categories. For example, Figure 12b and Figure 12c are coded as "oval" and "bar with rounded ends," respectively, and belong to two different categories. However, the perceptual boundary between these categories may become unclear as the shape is progressively elongated or compressed.

It is also difficult to exhaustively represent all possible feature variants when the labeling system is restricted to a limited set of basic categories such as circles, bars, and polygons. For example, Figure 12d represents a variant of a circle. Numerous visually different features may still be coded within the same "circle" category, making it impractical to collect examples of every possible variant for model training.



(a) A broken circle or four arcs



(b) Ellipse



(c) A rounded bar



(d) A circle with a knobbly perimeter

5.2 How Human Makes Judgements

A closer examination of the labeling process suggests that object classes should be defined according to common human visual perception. The trained network is intended to assist

footwear examiners, consistency between human interpretation and network labels is important. For this reason, the classification labels should align with human cognition in this study.

A ventral pathway is a pathway that process information that leads to recognition and identification of objects. Visual information is processed hierarchically along the ventral visual pathway. When human sees an object, the eye-seeing signals first reach V1 area, where basic features such as edges, orientations, and contrast are detected. These signals then travel through the nervous system, they are reorganized in a way so that they are received as an objects eventually. Gestalt psychologists provide principles that explain how visual system reorganize. According to Gestalt principle of relatability[19] and closure[20], the broken circle (Figure 12a) is perceived as a circle because the curvatures are compatible, consequently the missing contour can be interpolated and integrated into a continuous contour, and is perceived as a circle. Even larger gap can be filled that a non-existing object is perceived, the most famous example is the Kanizsa triangle[21] (Figure 13). In Figure 13, although a triangle whose edges are not physically present, human perceives the boarder of a invisible triangle from the incomplete shapes.

Neuroscientific studies suggests that there are the connected neurons in the primary cortex area (V1 area) compare their input, and they respond to illusory edges in a similar way of responding to a real line.

In addition to the coherent perception, human also weighs global contour organization and local features in the judgement. Loffler et al.[22] and Wilkinson et al.[23] studied smooth wave-like

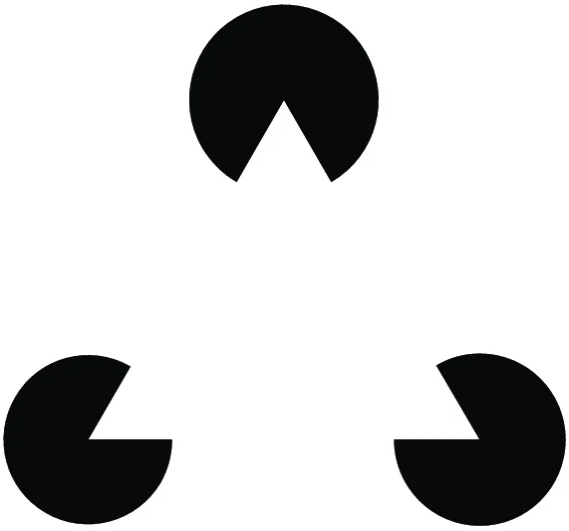
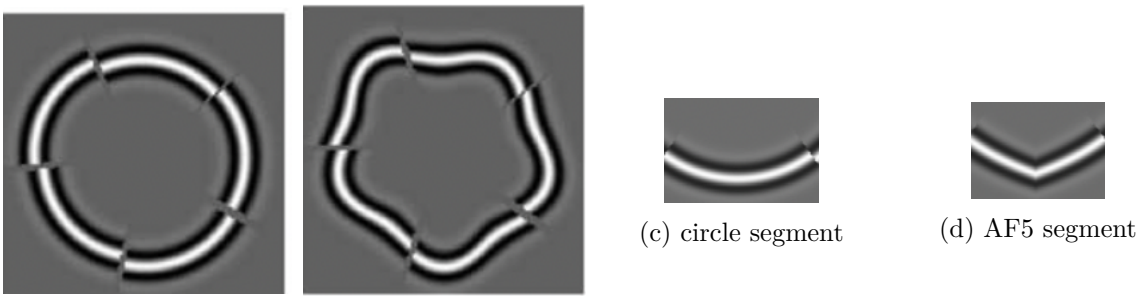


Figure 13: Kanizsa triangle

deformations of circles (radial frequency change Figure 14a Figure 14b). They found observers can detect very small shape changes, but when the contour is disrupted or rearranged, performance drops strongly. Wilson[24] studied controlled deformations that gradually transform circles into polygons (angular frequency change, Figure 14c Figure 14d), they found jittering the contour has little effect, and performance on single angle is close to performance on whole shape. Hence, curved shapes rely more on global contour integration, whereas shapes with corners rely more on local angle information. Weights of global and local information are assigned differently for integration depending on shape type when perceiving a shape.

5.3 How the Problem Will be Processed

Although visual networks have been developed for various purposes, their ability to recognize footwear features has not been thoroughly investigated. Footwear features present a particular



(a) circle jittering (b) RF5 jittering (c) circle segment (d) AF5 segment

Figure 14: Radial and angular deformations[23] [24]
(a)-(b) jittering impairs discrimination suggesting RF (radial frequency) perception depends n glnju9 obal contour arrangements
(c)-(d) single angle has little effect on discrimination suggesting AF (angular frequency) relies more on local features

challenge because many of them can be represented as basic geometric elements, while human perception of these elements is not always determined by strict geometric definitions. This makes the task different from common transfer-learning applications. CNNs incorporate several principles inspired by biological visual processing, while attention-based models focuses on attributing attention to more relevant features cross image regions. Although this is not a direct model of human attention, the adaptive weighting of relevant information is conceptually related to selective attention in the human visual system. Therefore, we will approach by examining whether visual networks can recognize geometric footwear features in a manner consistent with human perception. Specifically, we investigate their ability to extract basic geometric elements, the effects of backbone depth and architectural characteristics on feature representation, and whether networks trained on predefined shape categories can recognize and classify shapes formed through Gestalt perceptual organization.

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